

Teorema de Abel

Teorema 1 (de Abel). Sea $f(x) = \sum_n a_n x^n \in A[[x]]$ una serie formal de potencias con coeficientes en $\mathbb{C}$ (o $\mathbb{R}$) tal que $f(x_0)$ converge

$\implies \forall x : |x| < |x_0| : f(x)$ converge uniformemente.

Demostración: Como $\sum_n a_n x_0^n$ converge, se tiene que $a_n x_0^n \xrightarrow{n \rightarrow \infty} 0$ y, por tanto, $\exists C > 0 : \exists N_0 \in \mathbb{N} : \forall n \geq N_0 : |a_n x_0^n| \leq C$. Sea $r \in \mathbb{R}$ y $x \in \mathbb{C}$ (o $\mathbb{R}$) tal que $|x| \leq r < |x_0|$

$$\implies |a_n x^n| = |a_n| |x|^n \leq |a_n| r^n \frac{|x_0|^n}{|x|^n} \leq C \left(\frac{r}{|x_0|} \right)^n =: M_n.$$

Aplicando el criterio de Weierstrass, tenemos que $\sum_n a_n x^n$ converge uniformemente en cada disco cerrado $\overline{D}(0, r)$. ■

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